Angles and Measurement - Practice Answers

Part A | Angles

Each answer gives the result, a short line of working where it helps, and the reason for each angle fact.

A1 Points, lines, angles and notation

1 Y7 ICE-EM Ex 6A Q3

a AX, BX, CX, YX **b** $AC \parallel XY$ **c** B **d** CX
e A, B, C (AC and CB are the same line) **f** None, the two lines are parallel.

2 Y7 ICE-EM Ex 6B Q2

a concurrent **b** yes **c** no **d** ray PA and ray PK
e ray AP, ray AB, ray AO and ray AK

3 Y8 Cambridge Ex 6A Q5

a i $\angle DOE$ ii $\angle AOB$ iii $\angle DOE$ or $\angle AOB$ iv $\angle COD$
b i $AD \perp OC$ ii $OC \perp OD$

4 Y7 ICE-EM Ex 6B Q7

a $\angle BAD$, $\angle BCD$, $\angle ADC$, $\angle ADB$, $\angle CDB$, $\angle ABC$, $\angle ABD$, $\angle CBD$
b $\angle BAC$, $\angle CAD$, $\angle DAB$, $\angle ABC$, $\angle CBD$, $\angle DBA$, $\angle ACB$, $\angle BCD$, $\angle DCA$, $\angle ADB$, $\angle BDC$, $\angle CDA$

5 Y8 Cambridge Ex 6A Q10

a $(90 - x)°$ (angles in a right angle) **b** $(180 - x)°$ (angles on a straight line)
c $x°$ (vertically opposite angles) **d** $(360 - x)°$ (angles in a revolution)

A2 Measuring and classifying angles

1 Y7 ICE-EM Ex 6C Q3

a acute **b** obtuse **c** reflex **d** right

2 Y7 ICE-EM Ex 6C Q1

a 75° **b** 55° **c** 18° **d** 2° **e** 34°

3 Y7 ICE-EM Ex 6C Q5 The minute hand turns $360°$ in 60 minutes, so $6°$ each minute.

a 360° **b** 180° **c** 270° **d** 30° **e** 300° **f** 150° **g** 720° **h** 540°

4 Y7 Ch 6 Review Q4

$\angle AOC = 90° - 32° = 58°$ (adjacent angles at O). Reflex $\angle AOC = 360° - 58° = 302°$ (revolution at O).

5 Y7 ICE-EM Ex 6C Q4 Each hour mark on the clock is $360° \div 12 = 30°$ apart.

a 180° **b** 90° **c** 30° **d** 60° **e** 120° **f** 120° **g** 60° **h** 150°

6 Y7 Ch 6 Challenge Q7

The minute hand turns $6°$ each minute and the hour hand turns $30° \div 60 = 0.5°$ each minute. After t minutes the angle between them is $6t - 0.5t = 5.5t$ degrees. So $5.5t = 90$, and $t = 16\frac{4}{11}$ minutes.

A3 Angles at a point

1 Y7 ICE-EM Ex 6D Q5

e $\angle AOB = 180° - 30° - 100° = 50°$ (straight angle at O)
k $\angle AOB = 180° - (50° + 40° + 30° + 20° + 10°) = 30°$ (straight angle at O)
l $\angle AOB = 360° - (80° + 70° + 60° + 50° + 40° + 30°) = 30°$ (revolution at O)

2 Y7 ICE-EM Ex 6D Q7

g β = 50° (straight angle at W: $180° - 130°$); α = 50° (straight angle at W: $180° - 50° - 80°$)
h α = 70° (vertically opposite angles at M); β = 110° (straight angle at M); γ = 30° (straight angle at D: $180° - 150°$); θ = 110° (straight angle at D: $180° - 40° - 30°$)

3 Y8 Cambridge Ex 6A Q8 Nine people share the $360°$ at the centre.

a $360° \div 9 = 40°$ **b** 10 equal shares: $2 \times 36° = 72°$
c 12 equal shares: $4 \times 30° = 120°$ **d** 18 equal shares: $10 \times 20° = 200°$

4 Y7 ICE-EM Ex 6D Q8

c $2\theta = 40°$ (vertically opposite angles at V), so θ = 20°
g θ = 120° (straight angle EG: $180° - 60°$); below EG, $3\gamma + \gamma + 60° = 180°$ (straight angle EG; the angle next to γ is $60°$, vertically opposite), so $4\gamma = 120°$ and γ = 30°
h $\alpha + 50° + \alpha = 180°$ (straight angle at K), so $2\alpha = 130°$ and α = 65°

5 Y8 Cambridge Ex 6A Q13

a $(a + 50) + (a - 90) = 180$ (angles on a straight line), so $2a - 40 = 180$ and $a = 110$.
b The angles are $160°$ and $20°$, so the $(a + 50)°$ angle should be the larger looking angle.

6 Y8 Cambridge Ex 6A Q14

e $a + a + 3a + 4a + 3a = 360$ (angles in a revolution), so $12a = 360$ and $a = 30$
f $(7a + 21) + (3a - 11) = 180$ (angles on a straight line), so $10a + 10 = 180$ and $a = 17$

A4 Transversals and parallel lines

1 Y8 Cambridge Ex 6B Q2

a $\angle BCH$ **b** $\angle ABE$ **c** $\angle GCB$ **d** $\angle BCH$
e $\angle FBC$ **f** $\angle GCB$ **g** $\angle FBC$ **h** $\angle DCG$

2 Y7 ICE-EM Ex 6E Q7

d α = 42° (alternate angles, $RS \parallel TU$)
g θ = $180° - 116° = 64°$ (co-interior angles, $DG \parallel EF$)
k β = 69° (alternate angles, $CA \parallel BD$)

3 Y7 ICE-EM Ex 6E Q8

f α = 71° (corresponding angles, $AC \parallel BX$); β = 50° (alternate angles, $AC \parallel BX$)
h γ = 43° (alternate angles, $CB \parallel LM$); θ = 76° (alternate angles, $CB \parallel LM$)

4 Y8 Cambridge Ex 6B Q7

b The angle at the lower vertex inside the zigzag is $40°$ (alternate angles on parallel lines), so $a = 360 - 40 = 320$ (angles in a revolution).
e $a = 31$ (alternate angles on parallel lines; the $65°$ is not needed).
i $82 + (a + 65) = 180$ (cointerior angles on parallel lines), so $a = 33$.

5 Y8 Cambridge Ex 6B Q9

a $\angle AOB = (180 - a)°$ (cointerior angles on parallel lines)
b The angle at B between BO and the arrowed line is $(360 - a)°$ (angles in a revolution), and $\angle AOB$ is equal to it (alternate angles), so $\angle AOB = (360 - a)°$.
c $\angle AOB + b + a = 180$ (cointerior angles on parallel lines), so $\angle AOB = (180 - a - b)°$.

A5 Further problems and proving lines parallel

1 Y7 ICE-EM Ex 6G Q1

a $AB \parallel CZ$ (alternate angles are equal)
c $HJ \parallel KM$ (co-interior angles are supplementary: $75° + 105° = 180°$)
f $DE \parallel GF$ (alternate angles are equal). DG is not parallel to EF, since $75° \neq 70°$.

2 Y7 ICE-EM Ex 6F Q1

c α = $180° - 70° - 40° = 70°$ (straight angle at B); θ = $70° + 40° = 110°$ (alternate angles, $AC \parallel DE$)
e α = 70° (alternate angles, $AB \parallel JK$); β = 30° (alternate angles, $AB \parallel JK$); γ = $180° - 70° - 30° = 80°$ (straight angle at C)
h α = 90° (alternate angles, $PQ \parallel AB$); β = 60° (alternate angles, $RS \parallel AB$); γ = $90° - 60° = 30°$ (straight angle at O)

3 Y7 ICE-EM Ex 6F Q4

d $\gamma + 10° + 130° = 180°$ (co-interior angles, $FI \parallel GH$), so γ = 40°
h $\gamma + \gamma = 180°$ (co-interior angles, $LM \parallel NO$), so γ = 90°; $\theta + 40° + \theta = 180°$ (co-interior angles, $LM \parallel NO$), so θ = 70°

4 Y7 ICE-EM Ex 6G Q2

b $AB \parallel CD$ (corresponding angles are equal: both $50°$). Hence β = 35° (corresponding angles, $AB \parallel CD$).
d $AB \parallel CD$ (alternate angles are equal: both $40°$). Hence γ = 30° (alternate angles, $AB \parallel CD$).

5 Y7 ICE-EM Ex 6F Q2

b $\angle FQR = 180° - 25° - 18° = 137°$ (straight angle at Q), so $\angle POQ = 137°$ (corresponding angles, $PO \parallel FG$)
d $\angle POY = 160°$ (alternate angles, $XP \parallel OY$); $\angle QOY = 150°$ (alternate angles, $ZQ \parallel OY$); $\angle POQ = 360° - 160° - 150° = 50°$ (revolution at O)

A6 Triangles

1 Y8 Cambridge Ex 6C Q1

a right-angled triangle **b** isosceles triangle **c** acute-angled triangle **d** equilateral triangle
e obtuse-angled triangle **f** equilateral triangle **g** isosceles triangle **h** scalene triangle

2 Y8 Cambridge Ex 6C Q7

d $a + 60 = 110$ (exterior angle of a triangle), so $a = 50$
e The base angles are $(180 - 100) \div 2 = 40$ each (isosceles triangle), so $a = 180 - 40 = 140$ (angles on a straight line)
f The angle inside the triangle at the crossing is $180 - 115 = 65$ (angles on a straight line), so $a = 180 - 60 - 65 = 55$ (angle sum of a triangle)

3 Y8 Cambridge Ex 6C Q11

a Isosceles, since OA and OB are radii, so they are equal in length. **b** $\angle OAB$ and $\angle OBA$
c 30° **d** $180° - 2 \times 36° = 108°$ **e** $(180° - 100°) \div 2 = 40°$

4 Y8 Cambridge Ex 6C Q9

a The angle of the triangle at the right-hand vertex is $55°$ (alternate angles on parallel lines), so $a = 180 - 70 - 55 = 55$ (angle sum of a triangle)
b $a + 35 + 85 = 180$ (cointerior angles on parallel lines), so $a = 60$
c $74 + 81 + a = 180$ (cointerior angles on parallel lines), so $a = 25$

5 Y8 Cambridge Ex 6C Q10

c $\angle ACB = \angle ABC = 72°$ (isosceles triangle, $AB = AC$); in $\triangle ADC$, $a + 90 + 72 = 180$ (angle sum of a triangle), so $a = 18$

e $\angle ACB = 49°$ (vertically opposite angles); $\angle CAB = \angle CBA = a°$ (isosceles triangle, $CA = CB$); $a + a + 49 = 180$, so $a = 65.5$

f $\angle ACB = 180° - 150° = 30°$ (angles on a straight line); $a + 90 + 30 = 180$ (angle sum of $\triangle ABC$), so $a = 60$

A7 Quadrilaterals

1 Y8 Cambridge Ex 6D Q3

a $a = 180 - 76 = 104$ (cointerior angles on parallel lines); $b = 76$ (opposite angles of a parallelogram)

c $a = 180 - 52 = 128$ (cointerior angles on parallel lines)

2 Y8 Cambridge Ex 6D Q9

a true **b** false **c** true **d** true **e** false **f** true

3 Y8 Cambridge Ex 6D Q7

a $5x = 95$ (a kite has one pair of equal opposite angles), so $x = 19$

b $95 + 2x + x + 85 = 360$ (angle sum of a quadrilateral), so $3x = 180$ and $x = 60$

c $4x + 3x + 2x + x = 360$ (angle sum of a quadrilateral), so $10x = 360$ and $x = 36$

4 Y8 Cambridge Ex 6D Q10

No. The interior angles of a quadrilateral add to $360°$, and two reflex angles would add to more than $360°$ on their own.

5 Y8 Cambridge Ex 6D Q6

a The reflex angle inside the shape is $360 - 40 - 85 - 27 = 208$ (angle sum of a quadrilateral), so $a = 360 - 208 = 152$ (angles in a revolution)

c The angle inside the shape at the top is $180 - 74 = 106$ (angles on a straight line); the reflex angle inside is $360 - 106 - 17 - 22 = 215$ (angle sum of a quadrilateral); so $a = 360 - 215 = 145$

d The angle of the parallelogram at the top right is $180 - 106 = 74$ (cointerior angles on parallel lines), so $a = 74$ (vertically opposite angles)

e The diagonals of a rhombus bisect its angles, so the top-left angle is $2 \times 31 = 62$ and the bottom-left angle is $180 - 62 = 118$ (cointerior angles); $a = 118 \div 2 = 59$

Angles and Measurement - Practice Answers

Part B | Measurement

B1 Units and conversions

1 Y7 ICE-EM Ex 9A Q2

e 2.7 cm **h** 190 cm **k** 6340 m **l** 0.2 km **n** 2.901 km **s** 6.25 mm

2 Y7 ICE-EM Ex 9B Q6

1 dm = 10 cm, so 158 cm = 15.8 dm.

3 Y7 ICE-EM Ex 9A Q9

1 km = 1000 m, so there are $1000 \div 20 = 50$ gaps. There is one more marker than gaps: 51 markers.

4 Y7 Ch 9 Challenge Q11

From the first stop to the third stop there are 2 equal gaps, so each gap is 300 m. Nine stops have 8 gaps: $8 \times 300 =$ 2400 m.

5 Y7 Ch 9 Challenge Q10

In each row the plants run from 25 cm to 9.75 m, a span of 9.5 m, which is $9.5 \div 0.5 = 19$ gaps. So each row has 20 plants, and there are $12 \times 20 =$ 240 plants.

6 Y7 Ch 9 Review Q22

1 mile = $1760 \times 3 \times 12 = 63\,360$ inches = $63\,360 \times 2.54 = 160\,934.4$ cm, so a mile is 1.609 344 km.

B2 Perimeter

1 Y7 ICE-EM Ex 9D Q1

d 52 **e** 50 **f** 80

2 Y7 ICE-EM Ex 9D Q5

d $4a + 6b$ **e** $8a + 3b$ **f** $14x + 12y$

3 Y7 ICE-EM Ex 9D Q8

The horizontal steps add to the 10 m across the bottom, and the vertical steps add to the 10 m up the side. So the perimeter is $2 \times 10 + 2 \times 10 =$ 40 m.

4 Y7 Ch 9 Challenge Q14

The trick: the horizontal steps add to the 16 m across the bottom, and the vertical steps add to the 10 m up the side. So the perimeter is $2 \times 16 + 2 \times 10 =$ 52 m.

5 Y7 Ch 9 Challenge Q18

With whole-number side lengths, the rectangles are 1 m by 64 m (perimeter 130 m), 2 m by 32 m (68 m), 4 m by 16 m (40 m) and 8 m by 8 m (32 m). A square has the smallest perimeter (32 m).

6 Y7 Ch 9 Challenge Q19

The length and width add to $108 \div 2 = 54$ m. For example, 10 m by 44 m gives 440 m^2, 20 m by 34 m gives 680 m^2, and 27 m by 27 m gives 729 m^2. The closer the two sides are, the larger the area. A square has the largest area (729 m^2).

B3 Area of rectangles and composite shapes

1 Y7 ICE-EM Ex 9E Q4

a $5 \times 0.85 = 4.25$ m^2 **b** $0.8 \times 2 = 1.6$ km^2

2 Y7 ICE-EM Ex 9E Q7

a area 42 cm^2 **b** width 3 cm **c** length 5 m **d** width 12 km

3 Y7 ICE-EM Ex 9F Q5

The four rectangles cover $4 \times 5 \times 1 = 20$ m^2, but the four 1 m by 1 m overlaps are counted twice. Area $= 20 - 4 = 16$ m^2.

4 Y7 ICE-EM Ex 9F Q2

d $7 \times 9 - 1 \times 3 = 60$ m^2

f $6 \times 6 + 1 \times 3 + 1 \times 1 = 40$ cm^2

h $8 \times 6 - 6 \times 2 - 2 \times 2 = 32$ m^2 (the cut-out is a 6 m by 2 m rectangle and a 2 m by 2 m square)

5 Y7 ICE-EM Ex 9E Q15

Total area $= 8 \times 5.4 + 2.7 \times 8.9 + 3 \times 13.3 = 43.2 + 24.03 + 39.9 = 107.13$ $m^2 = 1\,071\,300$ cm^2.

So 1 071 300 tiles are needed, and the cost is $1\,071\,300 \times \$4.50 = \$4\,820\,850$.

B4 Areas of triangles and parallelograms

1 Y7 ICE-EM Ex 9G Q1

c $\frac{1}{2} \times 13 \times 4 = 26$ cm^2 **d** $\frac{1}{2} \times 14 \times 12 = 84$ mm^2

2 Y7 ICE-EM Ex 9G Q14

If the height is h cm, the base is $2h$ cm, so $2h \times h = 18$, $h^2 = 9$ and $h = 3$.

Height 3 cm, base 6 cm.

3 Y7 ICE-EM Ex 9G Q15

$8 \times 4 - 2 \times \frac{1}{2} \times 2 \times 4 = 32 - 8 = 24$ cm^2 (or base $XB = 6$ cm and height 4 cm: $6 \times 4 = 24$)

4 Y7 ICE-EM Ex 9G Q7

b $8 \times 3 + \frac{1}{2} \times 8 \times 3 = 24 + 12 = 36$ km^2

c $6 \times 12 + \frac{1}{2} \times 5 \times 12 = 72 + 30 = 102$ cm^2

d $\frac{1}{2} \times 8 \times 15 + \frac{1}{2} \times 15 \times 6 = 60 + 45 = 105$ mm^2

5 Y7 ICE-EM Ex 9G Q8

a Four triangles, each $\frac{1}{2} \times 12 \times 5 = 30$ m^2: $4 \times 30 = 120$ m^2

c $\frac{1}{2} \times 10 \times 5 - \frac{1}{2} \times 2 \times 2 - \frac{1}{2} \times 3 \times 4 = 25 - 2 - 6 = 17$ m^2

d $(3 + 7 + 3) \times (3 + 5 + 3) - \frac{1}{2} \times 7 \times 5 = 143 - 17.5 = 125.5$ m^2

B5 Special quadrilaterals

1 Y8 Cambridge Ex 3D Q3

d rhombus, $\frac{1}{2} \times 5 \times 3 = 7.5$ cm^2 **h** kite, $\frac{1}{2} \times 20 \times 30 = 300$ mm^2

k trapezium, $\frac{1}{2} \times 5 \times (9 + 4) = 32.5$ m^2 **l** trapezium, $\frac{1}{2} \times 16 \times (20 + 50) = 560$ mm^2

2 Y8 Cambridge Ex 3D Q7

Area $= \frac{1}{2} \times 8 \times 14.5 = 58$ m^2, so the cost is $58 \times \$20 = \$1160$.

3 Y8 Cambridge Ex 3D Q11

No. The diagonals are not given. Use the formula for a parallelogram, $A = bh$, since we already know these lengths: $A = 10 \times 8 = 80$ cm^2.

4 Y8 Cambridge Ex 3D Q10

a $A = 2x^2$ **b** $A = 3ab$ (diagonals $2a$ and $3b$) **c** $A = 8a^2$

5 Y8 Cambridge Ex 3D Q5

The marks show the top rod and both halves of the cross rod are 30 cm. So the diagonals are $30 + 30 = 60$ cm $= 0.6$ m and $30 + 60 = 90$ cm $= 0.9$ m.

Area $= \frac{1}{2} \times 0.6 \times 0.9 = 0.27$ m^2

6 Y8 Cambridge Ex 3D Q8

Let the shorter side be x cm, so the longer side is $3x$ cm. Then $\frac{1}{2} \times 2 \times (x + 3x) = 12$, so $4x = 12$ and $x = 3$.

The parallel sides are 3 cm and 9 cm.

Angles and Measurement - Practice Answers

Part C | Mixed challenge

1 Y7 Ch 6 Review Q7

b At the right-hand vertex, the arrowed ray makes $55°$ with the base (corresponding angles, parallel lines) and $90°$ with the slanting side (alternate angles, parallel lines). So α = $180° - 55° - 90° =$ 35° (straight angle).

d Use the dashed parallel line. The upper segment makes $180° - 123° = 57°$ with it and the lower segment makes $180° - 150° = 30°$ with it (co-interior angles, parallel lines). So α = $180° - 57° - 30° =$ 93° (straight angle).

2 Y7 Ch 9 Review Q26

a $\frac{1}{2} \times 3 \times 4 =$ 6 cm^2

b Using BC as the base, $\frac{1}{2} \times 5 \times AX = 6$, so $AX =$ 2.4 cm.

3 Y7 Ch 6 Challenge Q1

Draw a line through each corner parallel to the two arrowed lines and use alternate angles at each corner in turn. The first segment after the right angle makes $90°$ with the parallels. At the $110°$ corner the next segment makes $110° - 90° = 20°$. At the $115°$ corner the next segment makes $115° - 20° = 95°$. So the unknown angle is α = 95°.

4 Y7 Ch 9 Review Q8

a Perimeter $= 6 + 1 + 3 + 3 + 1 + 4 + 2 + 5 + 3 + 5 + 3 + 6 =$ 42 m

Area $= 6 \times 6 + 2 \times 7 + 1 \times 3 + 3 \times 5 = 36 + 14 + 3 + 15 =$ 68 m^2

b Perimeter $= 10 + 6 + 4 + 6 + 10 + 6 + 6 + 10 + 6 =$ 64 km

The figure is three right-angled triangles, each with shorter sides 6 km and 8 km. Area $= 3 \times \frac{1}{2} \times 6 \times 8 =$ 72 km^2

5 Y7 Ch 6 Challenge Q2

a Draw lines through the two middle corners parallel to the arrowed lines. At the left corner, α splits into $20°$ (alternate angles) and $\alpha - 20°$. At the right corner, $50°$ splits into $\alpha - 20°$ (alternate angles) and $30°$ (alternate angles). So $50 = \alpha - 20 + 30$, and α = 40°.

b In the same way, x splits into γ and $x - \gamma$, and $\beta = (x - \gamma) + \alpha$. So $x = \gamma + \beta - \alpha$.

6 Y7 Ch 6 Challenge Q3

Draw lines through the corners at x and α parallel to the arrowed lines. At the corner x, the upper segment makes $180° - \gamma$ with the parallel line (co-interior angles), so the middle segment makes $x - (180° - \gamma)$ with it. At the corner α, α splits into $x - 180° + \gamma$ (alternate angles) and β (alternate angles). So $\alpha = x - 180° + \gamma + \beta$, and $x = 180° - \gamma + \alpha - \beta$.

7 Y7 Ch 9 Challenge Q4

The grassed area is two 9 m tall blocks, 8 m and 4 m wide, joined by a 4 m wide strip that is $9 - 3 - 3 = 3$ m tall.

Area $= 8 \times 9 + 4 \times 9 + 4 \times 3 =$ 120 m^2.

Perimeter $= 2 \times (16 + 9) + 4 \times 3 =$ 62 m (the outside rectangle plus the four 3 m sides of the two gaps).

The outer edge of the path encloses $10 \times 11 + 6 \times 11 + 2 \times 5 = 186$ m^2, so the path has area $186 - 120 =$ 66 m^2.